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Bangladeshi Rice and Tea leaf disease detection using Deep Learning

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***Abstract* – In the heart of Bangladesh's agricultural landscape, where the cultivation of rice and tea supports both livelihoods and food security, the constant danger of crop diseases is a substantial obstacle to economic stability. This research project aims to develop innovative disease detection models specifically designed for rice and tea leaves in Bangladesh. The study will utilize deep learning with sophisticated image analysis methods. Early detection of diseases is essential due to the extensive intake of these basic food items. Additionally, the initiative seeks to advance precision agricultural techniques, decrease the reliance on pesticides, and improve food security. The MobileNetV3 Large model stood out with its remarkable accuracy of 93.20% and required minimum training time, even though it was trained on a very small dataset gathered from real-world scenarios. This project seeks to modernize the agricultural sector in Bangladesh and empower smallholder farmers by transferring technology, promoting sustainable farming practices, and aligning with government objectives. The project aims to utilise the capabilities of MobileNetV3 Large and include a small amount of field data to reduce farming losses and significantly contribute to the resilience and profitability of Bangladesh's important agricultural industry.**

Keywords* – *EfficientNetB7, MobileNetV3Large, ResNet101, Leaf, Tea, Rice, Disease detection, Deep learning, Transfer learning.

I. INTRODUCTION

The agricultural sector is vital to the country, especially in Bangladesh's fertile soil where the cultivation of rice and tea is a way of life. However, despite the beautiful scenery and growing plantations, there is a silent threat that never goes away which is illnesses that affect the same crops that the country depends on. Bangladesh, a country with a strong agricultural legacy and a sizable farming population, is faced with the ongoing problem of preserving staple crops like rice and the business that drives the country's economy the tea sector. With the goal of using machine learning, the pinnacle of contemporary technological power, to combat the persistent problem of diseases in rice and tea leaves, this thesis sets out on a transformative journey into the heart of Bangladesh's agrarian tapestry. The importance of developing agricultural methods cannot be overstated as the world struggles with the complexities of a rapidly changing climate and the need for sustainable practices.

The research presented here is motivated by two times dedications: first, to protect the customs and means of livelihood that are established in Bangladesh's agricultural past; and second, to welcome the possibility that modern technology can bring these customs into a more robust and technologically advanced future. The focal point is disease detection, which represents a pivotal point at which the combination of traditional methods and modern artificial intelligence can pave the way for increased agricultural yields, financial stability, and food security. This study aims to develop deep learning models that are specifically tailored to the difficulties of disease identification in rice and tea, the two main crops in Bangladesh. A detailed approach is necessary due to the complexity of agricultural ecosystems, where factors include soil conditions and climatic patterns. This research initiative to create models that can identify early indicators of diseases using a thorough investigation of machine learning algorithms. This will enable farmers to obtain accurate and timely information for efficient action by seeing the leaves.

The following is a summary of our work's main contribution:

- Own dataset contains 349 photos that were physically collected from the tea estates of Sreemangal.
- A transfer-learning model with the highest accuracy of 93.20 %.
- Combined disease detection of two popular food items in Bangladesh.

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II. LITERATURE REVIEW

This paper [1] use CNN models for automatically detecting and classifying diseases in tea leaves. They used a dataset of 860 images consisted of 8 classes, which included both healthy plants and plants with various diseases like Algal leaf spot, Gray blight, White spot, Brown blight, Red scab, Bud blight, and Grey blight. The authors trained and tested the model using 80% and 20% of the dataset, respectively. The model achieved an accuracy of 94.45% in identifying the tea leaf diseases. The authors suggest that their proposed model can be further improved by varying the number of layers and applying the transfer learning concept.

In this document [2] involves an integrated approach using IoT and machine learning for the early prediction and broadcasting of tea leaf disease. The process includes the collection of data on temperature and humidity through IoT devices, capturing images of tea leaves using a camera, and analyzing the images using image processing techniques. Then network-based classifier is applied to identify and classify the diseases based on morphological features such as shape, size, and color of the leaves. The data collected from IoT devices but there is no explicit mention about number of data. This approach aims to provide a cost-effective and efficient system for detecting and managing tea leaf diseases.

The authors [3] used a deep neural network model to detect tea leaf diseases. They used a specific architecture for the classifier, which consisted of several convolutional and pooling layers as well as fully connected layers. The authors created a labeled dataset of 5867 tea leaf images and achieved a high accuracy of 96.56% in disease detection. They compared the performance of their model with other existing methods and found that their model outperformed them.

Discussion about the use of deep feature-based identification [4] of rice leaf diseases using SVM. The dataset used in this study consists of 5932 images collected from the agricultural sites of Sambalpur and Bargarh districts in Odisha, India, of four types of rice leaf diseases, namely Bacterial Blight, Blast, Brown Spot, and Tungro. For classification, the authors employed various deep convolutional neural network models, including AlexNet, VGG16, VGG19, GoogleNet, ResNet18, ResNet50, ResNet101, InceptionV3, InceptionResNetV2, DenseNet201, and XceptionNet. They extracted deep features from specific layers of these models, such as fc6, fc7, and fc8, and used them for classification using SVM. They also applied transfer learning to these models. The accuracy of the deep feature plus SVM approach for identifying rice leaf diseases using ResNet50 as the deep feature model is 98.38%.

The article [5] employs an advanced deep convolutional neural network (CNN) known as LeNet to accurately categorize 80 different photos of tea leaf illnesses, such as blister blight, red scab, red leaf spot, and leaf blight. The CNN classifiers were assessed for accuracy using ROC analysis, which revealed good levels of accuracy across several disorders. The accuracy of identifying blister blight illness, for example, was 85%. Despite not achieving 100% accuracy, the LeNet model demonstrates good results, indicating potential for future enhancements. The Table provides the specific classification accuracy of the LeNet algorithm for various illnesses.

The publication explores the application of multi-task deep transfer learning in the identification of illnesses affecting rice and wheat leaves. The experimental findings demonstrate that the enhanced VGG16 model, coupled with the multi-task transfer learning approach, achieves precise identification of illnesses in rice and wheat leaves. The accuracy rate for rice is at 97.22%, while for wheat it reaches 98.75%. The multi-task model has superior performance compared to single-task learning approaches in terms of accuracy, with a 0.43% increase in accuracy rate. The dataset included in the studies comprises photos obtained from both field sources and publicly available photographs from Google. The enhanced VGG16 model surpasses traditional networks such as Resnet50 and Densenet121 in terms of accuracy and generalization skills.

The work explores the application of convolutional neural networks (CNNs) [7] in the prediction of rice leaf diseases. Two models were employed: a basic Convolutional Neural Network (CNN) and the InceptionResNetV2 model with transfer learning. The basic Convolutional Neural Network (CNN) attained an accuracy of 84.75% after 15 epochs, but the InceptionResNetV2 model achieved a higher accuracy of 95.67% after just 10 epochs. The InceptionResNetV2 model, when used with transfer learning, had the highest performance among all models, with an accuracy rate of 95.67%. The dataset utilized was obtained from Kaggle and Google pictures.

Image processing and feature extraction play a vital role [8] in the identification of diseases in tea leaves. The procedure includes the manipulation of pictures by cropping, scaling, and transforming them into threshold values. The collected characteristics are subsequently transmitted to the Artificial Neural Network Ensemble (NNE), which is trained via an ANN model. Negative correlation learning (NCL) is a method employed in neural network ensembles (NNE). The NCL model, which incorporates 10 neural networks with feature extraction, demonstrates the highest performance, obtaining an accuracy rate of 91%. The study utilized a dataset including 50 predetermined pictures.

A deep convolutional neural network (CNN) equipped with a multiscale feature extraction module [9] was employed to detect sick tea leaves. The "CIFAR10-quick model" has been enhanced with the "Alter-second model", resulting in an average identification accuracy of 92.5%. This accuracy surpasses that of typical machine learning methods and classical deep learning approaches. The technology eliminates the need for manual feature extraction and is capable of automatically extracting crucial information from illness photos. A total of 36 photos were processed and enhanced to train the deep Convolutional Neural Network (CNN) model.

Zhou et al. [13] introduced a rice disease detection method that combines Faster R-CNN and FCM-KM fusion. The approach demonstrated remarkable efficiency. While the detection accuracy of Faster R-CNN is commendable, its detection

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speed is sluggish, making it unsuitable for real-time applications. One-stage detection networks exhibit higher efficiency compared to two-stage networks, although at the cost of reduced accuracy.

In their study, Santanu Phadikar and colleagues [14] investigated a classification technique that utilises an automated system for the purpose of identifying illnesses in rice plants. The objective was to categorise various rice illnesses by extracting characteristics from the diseased region of the pictures under consideration. A technique known as fermi energy-centric segmentation was suggested to separate the disease-infected area of the picture from its backdrop. The symptoms of the affected area were identified based on attributes such as colour, location, and form, which were further analysed utilising algorithms. In order to reduce the complexity of the classifier and minimise the loss of information, significant characteristics are selected using Rough Set Theory (RST).

Tea is a widely consumed beverage globally, and in Bangladesh, [15] tea growing plays a crucial role. Several illnesses limit the optimal development of tea leaves, resulting in a decrease in tea produce. The proposed algorithm [16] effectively detects five types of persistent tea leaf diseases, including Red Rust, Red Spider, Thrips, Helopeltis, and Sunlight Scorching, with an average accuracy of 83%, thereby reducing global crop yield loss. The paper [17] identifies a pattern in infrared image gray distribution for tea disease, enabling the extraction of two characteristic parameters into a classifier for rapid detection. The algorithm includes canopy infrared thermal image classification, color identification, noise filtering, binarization, closed operation, and counting. [18] Tea is a widely consumed beverage globally, and in Bangladesh, the cultivation of tea is of great importance. Several illnesses impede the optimal development of tea leaves, resulting in a decrease in tea yield.

The author [19] employed a total of 300 photographs, which were partitioned into separate sets for both training and testing objectives. The simulation findings demonstrate a 99% accuracy for the blast-affected photos and a 100% accuracy for the normal pictures throughout the training stage. The accuracy of the testing phase is determined to be 90% and 86% for the photos of infected and healthy subjects, respectively. This study is to introduce a novel approach called attention-based depthwise separable neural network with Bayesian optimization (ADSNN-BO) for the identification and categorization of rice diseases using pictures of rice leaves [20]. Rice infections sometimes lead to a yield loss of 20-40% in the cultivation of rice and have a significant impact on the worldwide economy. The experimental findings indicate that their mobile compatible ADSNN-BO model obtains a test accuracy of 94.65%.

This study [21] identifies three frequent rice plant diseases: leaf smut, bacterial leaf blight, and brown spot diseases. The input consisted of clear photos of rice leaves showing the afflicted areas against a white backdrop. The dataset underwent essential pre-processing before being trained using several machine learning techniques, such as KNN , J48 , Naive Bayes, and Logistic Regression. Various processing of pictures and machine learning techniques have been developed to diagnose rice plant illnesses. The accuracy of plant disease detection using algorithms for machine learning depends on three key processes: feature segmentation, and feature extraction, and classification method. Artificial neural network approaches have demonstrated promising outcomes in the field of picture classifying. In recent studies, scientists have investigated the illnesses affecting mango [22], apple [23], tomato [24], rice [25] and wheat [26]. They have often employed fruits or leaves to diagnose illnesses based on the visuals. In addition, they have employed photographs with uniform backgrounds in specific situations.

The followings are some motivation of our work:

- The goal of this research paper is to contribute to the advancement of rice & tea quality detection methods in Bangladesh, and to provide useful information and recommendations for stakeholders in the rice & tea industry.
- Ensuring high-quality rice is essential to improve the overall food quality in Bangladesh and good quality tea leaves for the consumers.
- Developing a CNN-based model for rice & tea leaf disease detection can contribute to improving food characteristics, crop loss prevention, reducing chemical usage, sustainable agriculture and promoting public health.

III. RESEARCH METHODOLOGY

The focus of this research is the detection of diseases in rice and tea leaves in Bangladesh using machine learning techniques where the workflow start with data collection.

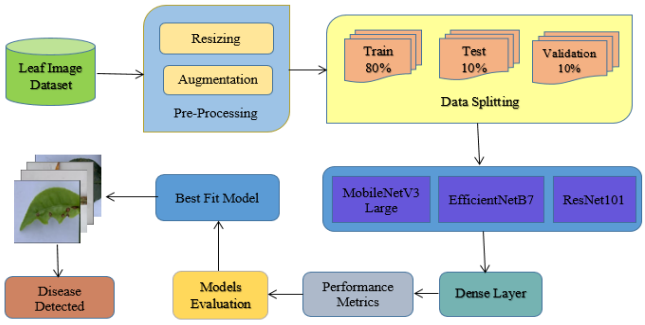


Figure 1. Research Methodology Steps

A. *Research Subject and Instrumentation*

Our study focuses primarily on Deep Learning Approaches and is titled "Bangladeshi Rice and Tea leaf disease detection using Deep Learning". The experiment's models were created using the Google Collaboratory's T4 GPU. The deep learning-based model was developed using the

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scikit-learn, pandas, and NumPy libraries. Python, a programming language, was generally utilized. We utilized Google Collaboratory to construct and simulate our models.

B. *Data Pre-Processing*

The tea data set was collected using a smartphone on the tea Estates of Sreemangal, Bangladesh and where the photographs were in different size. It was quite challenging to train and test the dataset. The current iteration of the dataset has images with a consistent resolution. As per the specifications of my assignment, we have transformed all the photos into a square shape. we standardized the resolution of the photos to 224 × 224. Initially, we performed picture cropping to eliminate any undesired things included in the photographs and lessen the size of the images. Furthermore, we used the RGB color mode to train our model.



Figure 2. Sample Pre-Process Data

C. *Data Separation*

Following the completion of the data augmentation step, the pre-processed dataset was segmented into a training set, a validation set, and a testing set. The pre-processed dataset was divided into a training set, validation set, and testing set in 80%, 10% and 10% sequentially. The deep CNN model was produced by employing the training set, which needed the use of transfer learning techniques in order to fine-tune a CNN model that has been trained in the past. Transfer learning may be illustrated by the process of taking a CNN model that has already been trained and modifying the model's final few layers so that it can be utilized for a new job. During the training phase, [11] we kept a close eye on how well the model performed with the validation set and adjusted a number of hyperparameters in order to make it work more effectively. The trained model was then evaluated using data that it had never been exposed to previously throughout the evaluation process. In general, our method consisted of amassing a large number of photographs of leaves, cleaning the data by means of data augmentation, and fine-tuning a pre-trained CNN model by means of transfer learning with the intention of identifying the many kinds of plums that are native to Bangladesh.

D. *Architecture of the Model*

EfficientNetB7:
EfficientNetB7, a constituent of the Efficient Net framework, is built with a distinctive and effective structure. The architecture consists of four convolutional layers, each utilizing common parameters for padding and activation. The model begins with 32 filters for the Conv2D variable and gradually increases to 512 filters in the following layers, which is very noteworthy. The rectified linear unit (ReLU) activation function is constantly utilized to activate every Conv2D layer. The development of EfficientNetB7 incorporates a well-balanced combination of depth, width, and resolution, resulting in an optimized and resource-efficient model that is very effective for a range of computer vision tasks.

MobileNetV3Large:
MobileNetV3Large, a member of the MobileNet family, is specifically designed to deliver exceptional performance in situations with limited resources. The model has a unique architecture that includes shared padding and activation parameters in its four convolutional layers. It starts with a Conv2D variable of 32 and ends with 256 filters. The Rectified Linear Unit (ReLU) activation function is consistently utilised for every Conv2D layer. The architecture of MobileNetV3Large focuses on inverted residuals and linear bottlenecks, achieving a trade-off between accuracy and efficiency. This makes it highly suitable for real-time applications on mobile and edge devices.

ResNet101:
ResNet101 is a convolutional neural network that use skip connections to address the issue of disappearing gradients by employing residual learning. By including parallel convolutional layers, this model effectively captures multi-scale properties that are essential for accurate picture categorization. The combination of batch normalization and regularization improves the efficiency of training and the ability to generalize. The novel design of this architecture guarantees strong performance in computer vision tasks.
Dense Architecture:
The base layer of neurons in the brain is referred to as the Dense Layer due to the fact that all the neurons in the layer above transmit information to it. The output generated by the convolutional layers is utilized by the Dense Layer to categories the processed pictures. The intricate structure of each model is meticulously depicted in a highly comprehensive manner in Table I. Every CNN model possesses an identical dense organizational structure.

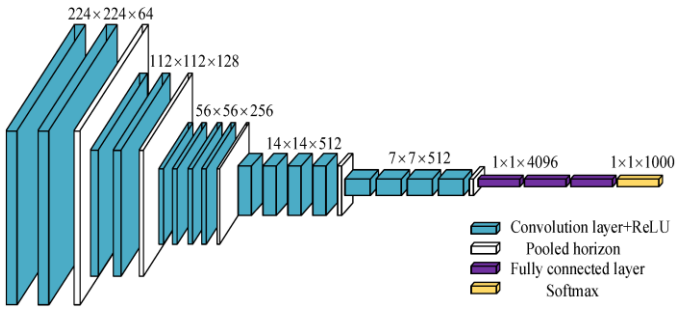


Figure 3. Model Architecture. [12]

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IV. MATHEMATICAL MODEL

A. Training, Testing and the Validation of the model

The primary objective of our inquiry is to utilise deep learning techniques to identify diseases in rice and tea leaves. We will employ a deep learning methodology employing smartphone photos. Our study in Python involved using Google's Tensor Flow and the Keras deep learning framework for image processing. The experiment's models were generated utilizing the GPU of Google Collaboratory. A deep learning model was developed using the scikit-learn, pandas, and NumPy libraries. We used Google Collaboratory to construct and simulate our models. The aggregate quantity of photographs is around 1776, with a distinction made between training, test, and validation data. We integrated the RGB color mode with the category mode. The proposed model showed excellent performance in the conducted investigation.

The use of a confusion matrix is necessary for the quantification of measurements. A confusion matrix, also known as an error matrix, is a table structure utilised in deep learning to illustrate the effectiveness of a supervised learning approach. The determination of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) is accomplished through the use of a confusion matrix.

- Accuracy: Accuracy is the most evident performance metric. The ratio of accurately predicted observational data to the total number of observations remains, in essence, unchanged.

$$Accuracy=(TP+TN)/(TP+FP+FN+TN)\ldots (1)$$

- Precision: The accuracy ratio evaluates the degree to which those measurements that were correctly predicted as positive align with all the expected positive studies. What is the number of persons who have remained unharmed, according to metrics.

$$Precision=TP/(TP+FP)\ \ldots..(2)$$

- Recall: Recall is the ratio of correctly predicted positive observations to all actual positives. It measures the ability of the model to capture all positive instances.

$$Recall=TP/(TP+FN)\ \ldots.(3)$$

- F1 Score: F1 score is the harmonic mean of precision and recall. It provides a balance between precision and recall, offering a single metric that considers both false positives and false negatives.

$$F1\ Score=2*(Recall*Precision)\ /\ (Recall*Precision)\ldots.. (4)$$

Model Analysis: In this research, three deep learning models are used on the rice and tea leaf disease dataset with particular parameters which are described in the earlier section. Three distinct models—MobileNetV3Large, EfficientNetB7, and ResNet101 are depicted below according to validation accuracy, test accuracy, and training duration. The MobileNetV3Large model achieves the greatest validation accuracy of 94.74%, followed by the EfficientNetB7 model at 91.67%, and the ResNet101 model at 89.91%. Regarding test accuracy, MobileNetV3Large has the highest score of 93.20%, followed by EfficientNetB7 with 88.40%, and ResNet101 with 86.80%. The duration of training varies across different models, with EfficientNetB7 requiring the most time at 408.86 seconds, followed by ResNet101 at 385.45 seconds and MobileNetV3Large at 373.18 seconds.

Table-1. Model's Accuracy Report

Serial	Algorithm	Validation Accuracy (%)	Accuracy (%)	Training Time (Sec)
1	MobileNetV3Large	94.74	93.20	373.18
2	EfficientNetB7	91.67	88.40	408.86
3	ResNet101	89.91	86.80	385.45

According to table 1, model 1 which is MobileNetV3Large outperforms the other models. Consequently, we have determined that MobileNetV3Large will be our optimal choice.

V. RESULT & ANALYSIS

A. Result Discussion

Table-2. MobileNetV3Large Classification Report

Label Name	Precision	Recall	F1_Score	Support
Rice Blast	0.94	0.91	0.93	68
Rice Brown Spot	0.87	0.87	0.87	30
Rice Shath Blight	0.91	0.98	0.94	49
Rice Tungro	0.94	0.89	0.91	36
Tea Bird Eye Spot	1.00	1.00	1.00	13
Tea Brown Blight	1.00	1.00	1.00	6
Tea Gray Blight	0.87	1.00	0.93	13
Tea Healthy Leaf	1.00	1.00	1.00	13
Tea Leaf Rust	1.00	0.87	0.93	15
Tea Red Spider	1.00	1.00	1.00	7
Accuracy			0.93	250
Macro avg	0.95	0.95	0.95	250
Weighted avg	0.93	0.93	0.93	250

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Table 2 summarizes the extensive assessment indicators of a classification model, providing insights into its performance across various labels. The precision, recall, and F1-score are painstakingly documented for each class, providing insight into the model's capacity to accurately detect and categorize cases. The model achieves flawless scores in Tea Bird Eye Spot, Tea Brown Blight, Tea Healthy Leaf, and Tea Red Spider, demonstrating remarkably high precision and recall values across these categories.

The model's proficiency in accurately predicting labels is indicated by its total accuracy of 93.20%. The use of macro and weighted averages reinforces a well-rounded and strong performance, highlighting the model's dependability in effectively managing many categories within the dataset.

For categorization and detection of road damage, we can recommend the ResNet50 model above all others because it outperforms them. For the ResNet50 model, we achieved the greatest accuracy of 97.15%. The ResNet50 model likewise has the top scores for sensitivity and specificity.

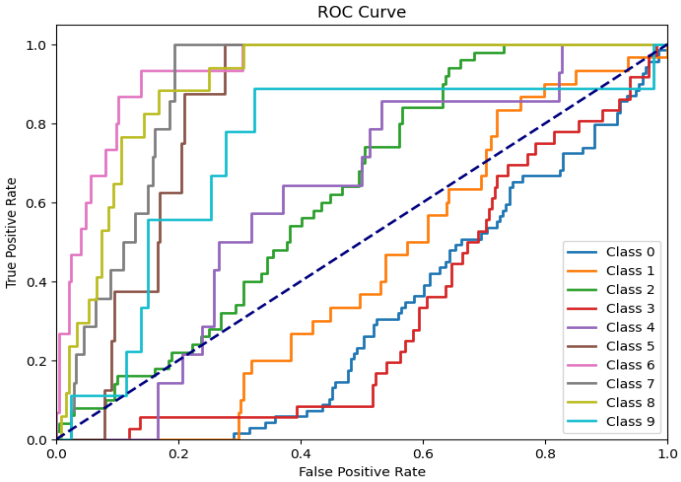


Figure 3. ROC Curve.

This ROC curve illustrates the performance of a multi-class classifier, with each class represented by a different color. The curves show the trade-off between the true positive rate and false positive rate for each class, indicating varying levels of classification accuracy.

Compared to other studies, this research achieves the highest accuracy. This work is the result of close collaboration among several researchers. Table 3 illustrates the connections between this study and previous research on various leaf disease detection tasks.

Table-3. Comparative Analysis

Work	Algorithm	Accuracy
Image analysis and detection of tea leaf disease using deep learning. [5]	CNN	85%
Tea leaf diseases recognition using neural network ensemble. [8]	NNF	91%
Identification of tea leaf diseases by using an improved deep convolutional neural network. [9]	CNN	85%
Bangladeshi Rice & Tea leaf disease detection using Machine Learning.	MobileNetV3 Large	93.20%

The table shows that my accuracy surpasses all previously mentioned works. By resizing images, MobileNetV3Large achieved a notable accuracy of 93.20%. This model effectively detect leaf disease.

VI. CONCLUSION AND FUTURE WORK

Three different models were used to identify diseases in rice and tea leaf crops in Bangladesh. Among these models, MobileNetV3Large demonstrated the highest level of accuracy, obtaining an impressive accuracy rate of 93.20%. The effective utilization of deep convolutional neural networks (CNNs) highlights their ability to precisely categorize illnesses. This study emphasizes the potential of deep learning and computer vision in identifying diseases, specifically in the agricultural setting. The impressive accuracy of 93.20% achieved by MobileNetV3Large demonstrates its reliability for practical use, particularly in improving crop management and decision-making procedures. The incorporation of a dataset consisting of 2540 photos underscores the significance of dataset magnitude and variety in effectively training precise models. Preprocessing methods and the careful evaluation of lighting conditions are crucial in improving the illness detection system. High-resolution plant photos obtained using digital cameras, scanners, or drones are preferred more and image preprocessing enhances picture quality and reduces noise, while image segmentation divides the image into segments to isolate the diseased area [11]. In addition to its applications in agriculture, this technology also plays a significant role in ecological research, conservation initiatives and the monitoring of biodiversity. Although the current accuracy is impressive, continuous study and enhancements are necessary to improve the illness detection system's effectiveness and adaptability in various environmental

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situations. This includes focusing on domain adaptation, constructing strong models, and ensuring scalability.

In the future, few works can be done to take the work further.

• The investigation of more complex strategies for domain adaptation, with the goal of improving the recognition system's generalization capabilities across a wide range of environmental variables and variations.

• Research into ensemble learning methods, which include mixing several different models in order to improve plum species recognition in terms of both its accuracy and its resilience.

• The use of new data modalities, such as spectrum or hyperspectral imaging, in order to extract features that are more complete and hence improve species classification.

• Investigating whether or not the created recognition system can be applied to additional plant species, hence broadening its scope of applicability to include broader agricultural and ecological settings.

• Taking into consideration the possibility of real-time development and deployment of the recognition system in field settings, with an emphasis on overcoming problems relating to computing efficiency and hardware requirements.

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